

Methods

XRO model

The XRO model consists of a nRO model for ENSO^{51,52} coupled to stochastic-deterministic models (that is, seasonally modulated first order autoregressive models) for the other climate modes^{53–55}:

$$\frac{d}{dt} \begin{pmatrix} \mathbf{X}_{\text{ENSO}} \\ \mathbf{X}_M \end{pmatrix} = \mathbf{L} \cdot \begin{pmatrix} \mathbf{X}_{\text{ENSO}} \\ \mathbf{X}_M \end{pmatrix} + \begin{pmatrix} \mathbf{N}_{\text{ENSO}} \\ \mathbf{N}_M \end{pmatrix} + \sigma_{\xi} \xi, \quad (1)$$

$$\frac{d\xi}{dt} = -r_{\xi} \xi + w(t), \quad (2)$$

where $\mathbf{X}_{\text{ENSO}} = [T_{\text{ENSO}}, h]$ and $\mathbf{X}_M = [T_{\text{NPM}}, T_{\text{SPMM}}, T_{\text{IOB}}, T_{\text{IOD}}, T_{\text{SIOD}}, T_{\text{TNA}}, T_{\text{ATL3}}, T_{\text{SASD}}]$ are state vectors of ENSO and other climate modes, respectively. This model allows for two-way interactions between ENSO and the other modes. Two indices are used to describe the oscillatory behaviour of ENSO^{52,56}. They consist of SSTAs averaged over the Niño3.4 region 170°–120° W, 5° S–5° N (T_{ENSO}) and thermocline depth anomalies averaged over the equatorial Pacific 120° E–80° W, 5° S–5° N (h), that is, the WWV index (with a constant factor of the area it covers). For other climate modes, we consider the SST indices of many climate modes (Supplementary Table 1) that have been shown to interact with ENSO, including the NPM^{12,38,57} and SPMM¹³ in the extratropical Pacific, the IOB^{14,58,59}, IOD^{15,43,60,61} and SIOD¹⁶ in the IO, and TNA^{17,62}, ATL3 (refs. 18,43,63,64) and SASD^{19,65} in the AO. We recognize the possibility of enhancing ENSO forecast skill by incorporating extra modes of variability, provided they directly interact with ENSO, show substantial memory extending beyond months and offer more sources of variability beyond the chosen eight.

The dynamics governing the state matrix $\mathbf{X}$ (consisting of ten variables) contains linear ($\mathbf{L}$), nonlinear ($\mathbf{N}$) and stochastic (ξ) terms. The linear dynamics contains four key submatrices, organized as follows:

$$\mathbf{L} = \begin{pmatrix} \mathbf{L}_{\text{ENSO}} & \mathbf{C}_1 \\ \mathbf{C}_2 & \mathbf{L}_M \end{pmatrix}, \quad (3)$$

where the linear operator submatrix $\mathbf{L}_{\text{ENSO}}$ describes the ENSO internal recharge–discharge dynamics^{52,66}, $\mathbf{L}_M$ represent the internal processes and interactions among the other climate modes; $\mathbf{C}$ are coupling submatrices, with $\mathbf{C}_2$ describing the impact of ENSO on other climate modes²⁹ and $\mathbf{C}_1$ describing the feedback of other modes on ENSO. To implement nonlinear dynamics associated with ENSO asymmetry, quadratic nonlinearities $b_1 T_{\text{ENSO}}^2 + b_2 T_{\text{ENSO}} h$ are incorporated into the SSTA equation of ENSO following ref. 51 and ref. 67, specifically, $\mathbf{N}_{\text{ENSO}} = [b_1 T_{\text{ENSO}}^2 + b_2 T_{\text{ENSO}} h, 0]$. These nonlinearities can be related to deterministic nonlinear ocean advection^{67,68}, as well as to atmospheric nonlinearity implicitly through the nonlinear SST–wind stress feedback^{69–71}. A local quadratic nonlinearity $b_3 T_{\text{IOD}}^2$ is also incorporated in the SSTA equation for the IOD following the recent insights from ref. 72 that IOD asymmetry is dominated by local nonlinear processes. The nonlinear terms for modes other than the IOD are set to zero given their observed smaller asymmetry and skewness⁷³ (Supplementary Fig. 5i,j,m–p), specifically, $\mathbf{N}_M = [0, 0, 0, b_3 T_{\text{IOD}}^2, 0, 0, 0, 0]$. Last, ξ is stochastic forcing due to weather and other high-frequency noise such as the Madden–Julian Oscillation and westerly wind bursts, which is approximated as red noise with decorrelation time scales of r_{ξ} and amplitudes of σ_{ξ} , respectively. Specifically, $w(t)$ in equation (2) denotes white noise with a Gaussian distribution $N(0, 2r_{\xi})$ ensuring that the variance of ξ is maintained at the unit level. We acknowledge the importance of the multiplicative (state-dependent) noise forcing on ENSO^{74,75}, however, accurately estimating the magnitude of the state-dependence remains a challenge with the observational data length.

Owing to the strong seasonal dependence of ENSO and other climate modes, we incorporate seasonality by estimating the operator matrix and nonlinear parameters as

$$\mathbf{L} = \mathbf{L}_0 + \sum_{j=1}^2 (\mathbf{L}_j^c \cos j\omega t + \mathbf{L}_j^s \sin j\omega t), \quad (4)$$

$$\mathbf{N} = \mathbf{N}_0 + \sum_{j=1}^2 (\mathbf{N}_j^c \cos j\omega t + \mathbf{N}_j^s \sin j\omega t), \quad (5)$$

where $\omega = 2\pi/(12 \text{ months})$, and the subscripts 0, 1 and 2 indicate the mean, annual cycle and the semi-annual components, respectively. The linear operator and nonlinear coefficients for the observations and CMIP simulations are estimated simultaneously by using multivariate linear regression and expressing the state vector tendency in equation (1) through a forward-differencing scheme following refs. 76,77 Compared to the conventional method, which estimates the annual cycle of operators by splitting the monthly data on each calendar month, our approach enables us to obtain the seasonal modulated operators without reducing sample size by a factor of 12. We emphasize that our approach constitutes the minimum number of degrees of freedom necessary to represent the seasonality. There are 50 parameters for each tendency equation of the ten variables in the system (except 60 for T_{ENSO} and 55 for T_{IOD}). To meet the rule of thumb for regression sample size (at least ten subjects per predictor)⁷⁸, 40–50 years of data are required to achieve a robust fit. The total number of parameters is 515, which is orders-of-magnitude fewer degrees of freedom than the AI models have in comparison, the last of which have substantially more than 100,000 free parameters⁸.

The noise parameters are determined from the residuals of the XRO fit. There are 20 total noise parameters: that is, a noise amplitude and decorrelation time scale for each of the ten variables in the system. The noise amplitude σ_{ξ} is estimated from the standard deviations of the residuals of the XRO fit. The decorrelation time scales are estimated as $r_{\xi} = -\ln(a_1)/\delta t$, where a_1 is the lag-1 autocorrelation of the residual of the XRO fit. The order of observed noise time scale r_{ξ}^{-1} is about 0.25–0.70 months.

The XRO builds on the legacies of the Hasselmann stochastic climate model capturing upper ocean memory in SST variability, and the recharge oscillator model for the oscillatory core dynamics of ENSO. As a multivariate dynamical system, compared with previous linear inverse models^{27,28,35,79,80}, the XRO offers an enhanced capability in representing the dynamics of ENSO (including recharge–discharge dynamics) and climate mode interactions, encompassing their seasonality and nonlinearity, which are of crucial importance in improving ENSO forecast skill. Moreover, the state vectors for linear inverse models are typically derived from the leading principal components truncated within the empirical orthogonal function space that, however, may not always represent physical processes.

nRO

To highlight the climate mode interactions, we compared the XRO model with a nRO, which is described as:

$$\frac{d}{dt} \mathbf{X}_{\text{ENSO}} = \mathbf{L}_{\text{ENSO}} \cdot \mathbf{X}_{\text{ENSO}} + \mathbf{N}_{\text{ENSO}} + \sigma_{\xi_{\text{ENSO}}} \xi_{\text{ENSO}}. \quad (6)$$

This model includes only processes internal to the tropical Pacific. The parameters for the nRO model are fitted separately.

Observational data

We use nine observational SST and three-dimensional ocean temperature datasets to account the uncertainties in estimating the SST in global oceans and subsurface state in the equatorial Pacific